LockNLeave - Smart IoT-Enabled Railway Luggage Storage System

Project Report

CuriousPARC 2026 – National Innovation Challenge

Team HackMates

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PROJECT SYNOPSIS

The Advanced Smart Locker Management System enhances the previously developed IoT-based locker prototype by introducing intelligent monitoring, multi-locker scalability, and automated management features. The upgraded system integrates ultrasonic-based occupancy detection, dynamic pricing algorithms, AI-based predictive usage analysis, and a centralized admin control panel. The system also includes SMS notification services, emergency unlock mechanisms, and automated session credit handling.

By combining IoT hardware, cloud-based data management, and intelligent software algorithms, the system aims to create a scalable locker infrastructure suitable for railway stations, metro stations, airports, and smart city environments.

Target Implementation: Indian Railways coaches, metro stations, airports, and public transport terminals.

PROBLEM STATEMENT

Train passengers face significant challenges regarding luggage security:

- **Theft Risk**: Unattended luggage in compartments vulnerable to theft, especially during station stops
- **No Secure Storage**: Current railway infrastructure lacks in-coach secure storage for valuables
- **Manual Systems**: Existing station cloakrooms use outdated token systems without digital tracking
- **Limited Accountability**: No digital trail for stored luggage
- **Passenger Anxiety**: Constant vigilance required, reducing travel comfort
- Existing smart locker prototypes lack **automatic occupancy detection**
- Most systems do not support **multi-locker scalability**
- No **predictive usage analysis** for resource planning
- Lack of **centralized admin control and analytics**
- No **dynamic pricing models based on demand**

Impact: Improved luggage security, reduced theft risk, and enhanced passenger convenience through automated smart locker infrastructure with real-time monitoring, scalable management, and data-driven optimization.

OBJECTIVES

Primary Objectives

1. Develop secure digital locker rental platform with automated access control
2. Implement dual access via QR codes and web portal
3. Create time-based auto-lock system with rental enforcement
4. Enable seamless digital payment integration
5. Implement ultrasonic sensors for smart availability detection
6. Develop a centralized admin dashboard for monitoring and control
7. Enable dynamic pricing based on demand and usage patterns
8. Implement predictive analytics to estimate locker demand
9. Design a scalable multi-locker architecture
10. Develop SMS notification and alert systems
11. Design a custom PCB for reliable locker hardware integration
12. Create an optimized mechanical locker structure for real-world deployment

Measurable Outcomes

- Achieve accurate locker occupancy detection using ultrasonic sensors
- Enable monitoring and control of multiple lockers through a centralized system
- Provide real-time locker status and transaction tracking via the admin panel
- Implement automated alerts and SMS notifications for security and session updates
- Demonstrate dynamic pricing based on locker usage and demand
- Analyze locker usage data to identify peak demand patterns
- Improve hardware reliability through custom PCB integration
- Ensure secure and fast locker access using PIN/QR authentication

SYSTEM ARCHITECTURE

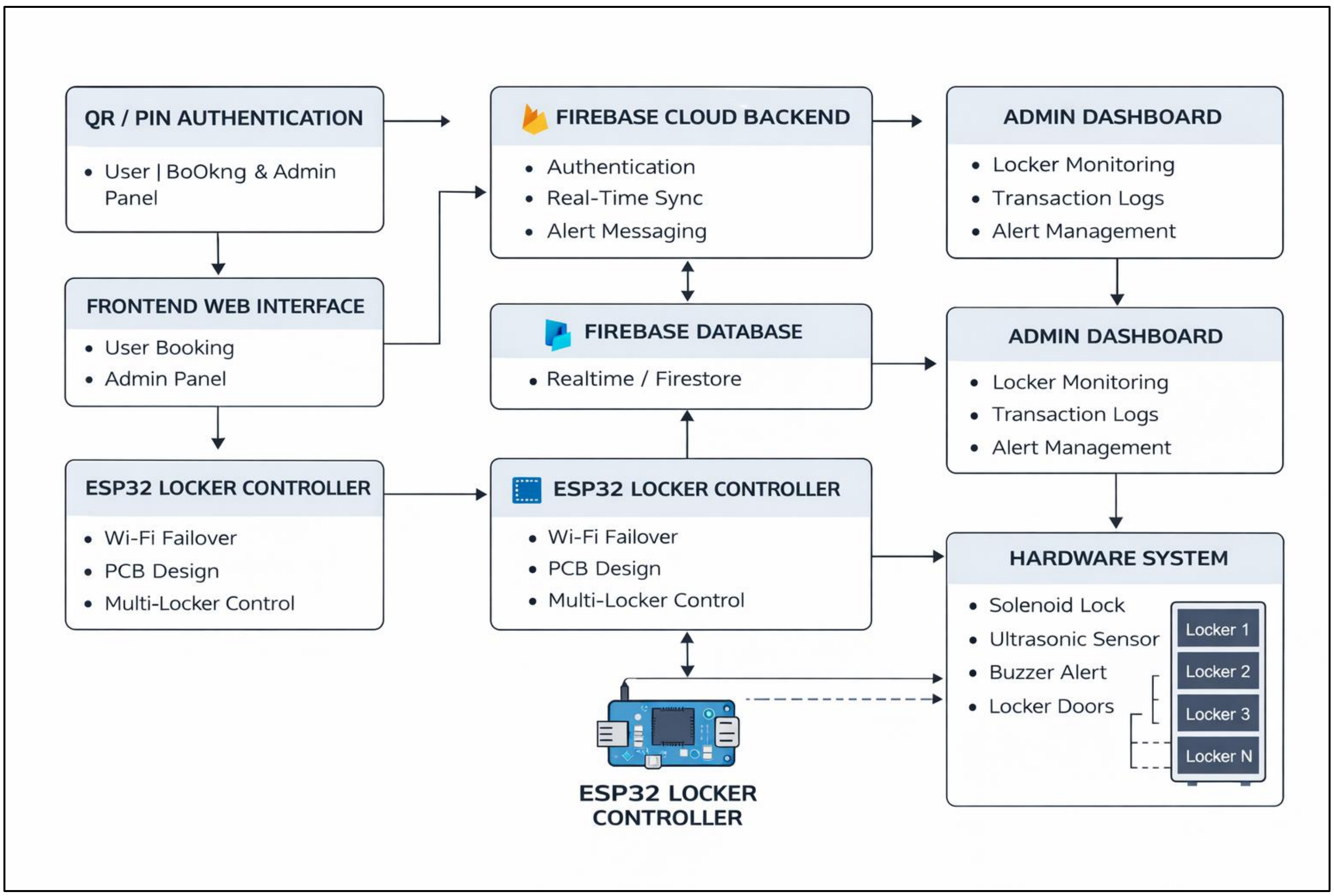
Key Components

- **Frontend:** Web interface for locker booking and admin monitoring
- **Backend:** Firebase for authentication and system communication

- **Database:** NoSQL database for user data and locker status
- **Hardware:** ESP32 controller with solenoid locking mechanism
- **Sensors:** Ultrasonic sensor for locker occupancy detection
- **Alert System:** Buzzer and SMS notifications for alerts
- **Security:** PIN and QR-based authentication

- **Expansion:** Multi-locker architecture for centralized control
- **Hardware Integration:** Custom PCB for stable circuit design

BLOCK DIAGRAM



METHODOLOGY

The development of the Advanced Smart Locker System was carried out through a series of structured stages integrating IoT hardware, cloud technologies, and intelligent software components.

- **System Design and Architecture:**
  The system architecture was designed to integrate IoT-based locker hardware with cloud-based services and a web interface. The design supports multi-locker management, secure user authentication, and centralized monitoring through an admin dashboard.

- **Frontend and Admin Panel Development:**
  A web-based interface was developed for users to book lockers and access them through PIN or QR authentication.

  An admin dashboard was also implemented to monitor locker status, manage transactions, and perform emergency unlock operations.

- **Cloud Backend Integration:**
  Firebase cloud services were used to manage authentication, real-time communication, and locker session data. The cloud backend enables secure data synchronization between the web application and the locker controller.

- **Database Management:**
  A NoSQL Firebase database was used to store user credentials, locker availability status, booking records, and transaction logs. This ensures fast data retrieval and real-time updates across the system.

- **Authentication and Access Control:**
  The system supports PIN and QR-based authentication for locker access. Each locker session is securely validated through the cloud backend before allowing the locker to unlock.

- **Hardware Implementation:**
  The locker system is controlled using an ESP32 microcontroller that communicates with the cloud database via Wi-Fi. The controller operates the solenoid lock mechanism and manages locker operations efficiently.

- **Sensor and Alert Integration:**
  Ultrasonic sensors were integrated inside lockers to detect luggage presence and determine locker occupancy. A buzzer alert system was added to notify events such as incorrect PIN attempts or session expiry.

- **PCB and Hardware Optimization:**
  A custom PCB design was developed to integrate the ESP32, sensors, power regulation circuits, and locking mechanisms into a compact and reliable hardware module.

- **Testing and Validation:**
  Extensive testing was performed to validate locker authentication, sensor accuracy, alert functionality, and real-time communication between hardware and the cloud system. The system was evaluated for reliability, response time, and overall user experience.

- **Data Analysis and Usage Monitoring:**

  Locker usage data such as booking time, duration, and occupancy patterns will be collected from the cloud database. This data will be analyzed to identify peak demand periods and usage trends, helping optimize locker allocation and improve system efficiency.

LITERATURE REVIEW

Fatima M. Inamdar (2025) – *IoT-Based Smart Locker System Using RFID*

- **Advantages:** Enabled IoT-based access and real-time locker monitoring.
- **Disadvantages:** Dependent on RFID tags; lacked advanced authentication security

Wang, Zhao & Li (2022) – IoT Smart Locker Infrastructure for Smart Cities

- **Advantages:** Supports cloud-based monitoring and scalable locker deployment.
- **Disadvantages:** Lacks advanced predictive analytics and real-time occupancy detection.

Zhang, Chen & Liu (2021) – *Secure Parcel Delivery Using Smart Lockers*

- **Advantages:** Provided secure data transmission and one-time access tokens.
- **Disadvantages:** Designed for fixed parcel lockers, not adaptable for railway use.

Kumar & Sharma (2020) – *IoT-Based Smart Locker System Using RFID*

- **Advantages:** Enabled real-time monitoring and IoT-based locker access.
- **Disadvantages:** Required physical RFID tags and lacked multi-layer authentication.

Patel & Mehta (2019) – *Cloud-Connected Locker Management System*

- **Advantages:** Offered centralized control and cloud-based scalability.
- **Disadvantages:** Focused mainly on software without strong hardware-level security.

Research Gap

Existing smart locker systems mainly focus on basic authentication methods such as RFID or parcel locker access. However, many systems lack intelligent features such as real-time occupancy detection, centralized cloud monitoring, predictive usage analytics, and scalable multi-locker architecture. There is also limited integration of IoT hardware with cloud-based admin control panels for automated locker management. This project aims to address these gaps by developing an advanced IoT-enabled smart locker system with cloud connectivity, sensor-based monitoring, and scalable locker infrastructure.

ADVANTAGES

For Passengers

- Enhanced security with PIN/QR authentication
- Easy locker booking through web interface
- Flexible rental duration options

- Digital transaction records and receipts
- Real-time locker availability detection

For Railways

- Additional revenue generation through smart locker services
- Reduced theft complaints and improved passenger safety

- Centralized locker monitoring through admin dashboard
- Real-time analytics and usage monitoring
- Scalable multi-locker infrastructure for large deployments

Technical Benefits

- IoT-based automated locker management system
- Cloud-based real-time monitoring using Firebase
- Sensor-based occupancy detection using ultrasonic sensors
- Alert system using buzzer and SMS notifications
- Improved hardware reliability using custom PCB design
- Data-driven insights for better locker deployment and infrastructure planning

LIMITATIONS

Technical

- Internet dependency (Mitigation: Offline mode with sync)
- Hardware integration complexity (Mitigation: Standard IoT protocols)
- Power supply needs (Mitigation: Battery backup with solar)
- QR code wear and tear (Mitigation: Durable materials, manual ID entry)
- Sensor accuracy limitations (Mitigation: Calibration and periodic testing)

Operational

- Initial infrastructure investment (Mitigation: Phased rollout)
- User adoption challenges for older passengers (Mitigation: Staff assistance)
- Regular hardware and sensor maintenance required (Mitigation: predictive alerts and system monitoring)
- Limited capacity during peak seasons (Mitigation: Dynamic pricing)

Security

- PIN theft risk (Mitigation: Biometric verification/OTP)
- Cyber-attack vulnerability (Mitigation: Encryption, security audits)
- Physical tampering possible (Mitigation: robust locks, sensors, tamper alerts, CCTV)

FUTURE SCOPE

The Advanced Smart Locker System has vast potential for expansion and improvement in the coming years. It can be integrated with AI-based predictive maintenance to automatically detect faults or usage trends in lockers. The system may also connect with Smart City and Digital Railway initiatives, enabling unified access across stations and transport modes.

Future versions can adopt green energy solutions like solar-powered lockers to promote sustainability. Adding data analytics will help track usage patterns and optimize locker placement for better passenger convenience. Features such as remote monitoring, voice assistance, and biometric authentication can further enhance security and user experience.

In the long run, this system can be scaled to metros, airports, and bus stations, supporting the goals of Digital India and Smart Transport Infrastructure.

IMPLEMENTATION ROADMAP

Phase 1 (Week 1–6): System design, project planning, development of web interface, and Firebase cloud database integration.

Phase 2 (Week 7–14): Hardware integration including ESP32 controller, solenoid lock mechanism, ultrasonic sensor, keypad, and buzzer alert system.

Phase 3 (Week 15–20): Development of admin dashboard, implementation of multi-locker architecture, and integration of locker monitoring features.

Phase 4 (Week 21–24): Data analysis of locker usage patterns, system testing, debugging, and final prototype validation.

CONCLUSION

The Advanced Smart Locker System enhances traditional locker infrastructure by integrating IoT technology, cloud-based monitoring, and intelligent management features. The system enables secure locker access using PIN and QR authentication while incorporating ultrasonic sensors for real-time occupancy detection and automated alerts for improved security. The integration of an admin dashboard allows centralized monitoring and management of multiple lockers, while data analysis of locker usage patterns provides insights for optimizing locker deployment and infrastructure planning. With its scalable architecture and smart monitoring capabilities, the system has the potential to support modern public transport environments such as railway stations, metros, airports, and bus terminals. This project demonstrates a practical step toward smart infrastructure and digital transport solutions.

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Field Hardware Setup

- All required hardware components for the IoT-Enabled Smart Locker System have been identified, designed, and integrated into a structured hardware architecture:
  - ESP32 (LilyGo T-Display) as the main controller
  - Solenoid lock with driver circuitry for controlled access
  - TFT display for system status and interaction
  - Ultrasonic sensor for occupancy detection
  - Regulated power supply module

- **PCB Design & Integration:**
  - Custom PCB design planned for compact and reliable integration of ESP32, power regulation, and driver circuits
  - Proper routing considered for voltage stability, current handling, and noise reduction
  - Modular PCB architecture designed to support multi-locker scalability

- **Embedded System Integration:**
  - ESP32 interfaced with TFT display for dynamic system feedback
  - Solenoid lock controlled via safe switching circuitry (relay/MOSFET-based)
  - Ultrasonic sensor enables real-time locker occupancy monitoring

- **System designed without dependency on physical input hardware:**
  - All control operations handled via cloud-based commands (Firebase → ESP32)

- **Prototype:**
  - Functional single locker prototype developed and tested
  - Hardware layout structured for expansion to multiple lockers

Software & Backend Development

- System built using a decoupled, event-driven architecture ensuring seamless synchronization between hardware and cloud systems

Technology Stack

- **Frontend & Dashboard:**
  - Next.js (React) for scalable web application
  - Zustand for state management and real-time updates
  - Tailwind CSS + Shadcn/UI for modern UI design
  - Recharts for analytics and visualization

- **Backend & Cloud:**
  - Firebase (Google Cloud Platform)
  - Realtime Database for hardware synchronization

- Firestore for business logic and pricing
    - JWT-based authentication for secure sessions

- **Firmware:**
    - ESP32 programmed using C++ (Arduino framework)
    - Secure communication implemented with cloud backend

Core Workflow

- User enters details on web interface → PIN generated on frontend → Stored in Firebase → Locker assigned → ESP32 validates → Locker unlocks
- **Session management:**
    - Auto session expiry
    - Real-time session tracking
    - Early termination support

PIN-Based Secure Access

- PIN generated at frontend and secured using SHA-256 hashing
- Only hashed values transmitted and verified
- Ensures user privacy and protection against data breaches

Real-Time Synchronization

- **Continuous communication between:**
    - Firebase ↔ ESP32 ↔ Web Dashboard
- **Enables:**
    - Instant locker status updates
    - Live session monitoring
    - Fast system response

Dashboard & Admin Panel

- Developed a centralized Admin Dashboard (System Nerve Center)
  Real-Time Monitoring
- Live tracking of locker availability and occupancy
- Instant reflection of hardware changes using Firebase listeners
- Monitoring of system health and connectivity

Remote Hardware Control

- Emergency override feature (force unlock via cloud command)
- Live visibility of active sessions and locker states

Data Analytics & Insights

- Visualization of:

    - Usage patterns
    - Peak demand periods
    - Revenue trends
- Helps in decision-making and optimization

Dynamic Pricing Engine

- Pricing tiers managed via Firestore
- Auto surge pricing based on demand threshold
- Simulation capability for pricing strategy testing

System Planning & Architecture

- **Complete system architecture finalized and implemented:**
    - Web Interface → Firebase → ESP32 → Locker Control
- **Features:**
    - Event-driven architecture
    - Multi-locker scalable design
    - Real-time bidirectional communication

- **Designed to handle:**
    - High user concurrency
    - Real-world deployment scenarios

Security & Blockchain Integration

- Implemented strong security mechanisms:

SHA-256 Hashing
- Used for:
    - Secure PIN storage
    - Data integrity verification

Blockchain-Inspired Hashing Layer

- Each locker event generates a hash linked to the previous event
- Creates a tamper-proof chain of records

System Security Features

- Role-Based Access Control (RBAC)
- JWT-based authentication
- Secure cloud communication
- Heartbeat protocol for detecting offline/tampered devices

Testing & Validation

- **Hardware Testing:**
    - Solenoid lock operation verified
    - Sensor-based occupancy detection tested

- **Backend Testing:**
    - Firebase read/write operations validated
    - Real-time synchronization confirmed

- **Load Testing (Apache JMeter):**
    - 10 users → Avg response: 96 ms, 0% errors
    - 20 users → Avg response: 92 ms, 0% errors
    - 50 users → Avg response: 95 ms, 0% errors
    - System demonstrated high reliability under load

- **End-to-End Testing:**
    - Booking → PIN → Validation → Unlock → Session end successfully executed

Current Status

- Fully functional integrated prototype developed
- Real-time cloud-hardware synchronization achieved
- Admin dashboard operational
- Security and blockchain mechanisms implemented

Work Remaining (Next Phase)

- Multi-locker physical deployment
- Mobile application development
- Integration with smart transport systems
- AI-based demand prediction and analytics enhancement

Final System Development & Integration

- Successfully developed and integrated the complete IoT-enabled Smart Locker System prototype.
- Implemented decentralized locker architecture where each locker operates independently using ESP32 controllers.
- Replaced traditional keypad-based authentication with TFT Touchscreen based Virtual PIN Entry System.
- Developed secure PIN generation and validation mechanism integrated with Firebase cloud services.
- Implemented real-time communication between ESP32 hardware, Firebase Realtime Database, and Web Application.

System Overview

The developed IoT-Enabled Smart Locker System provides secure and intelligent luggage storage using ESP32-based locker controllers, Firebase cloud infrastructure, and a web-based booking platform. The system enables users to reserve lockers online, receive secure virtual PIN credentials, and access lockers through a touchscreen interface. Real-time synchronization between cloud services and hardware allows centralized monitoring and management of multiple locker units.

Hardware Development

Designed and assembled a complete locker hardware prototype integrating cloud-connected embedded electronics and intelligent monitoring devices.

System Components:

- ESP32 Development Board
  - Acts as the central controller of the system.
  - Handles Firebase communication, PIN validation, sensor monitoring, and lock control.

- TFT Touch Display Module
  - Provides a virtual PIN entry interface.
  - Displays locker status, authentication messages, and system notifications.

- Solenoid Lock Mechanism
  - Provides secure locker locking and unlocking operations.
  - Activated only after successful PIN authentication.

- Reed Switch Door Monitoring Sensor
  - Detects locker door open and close status.
  - Sends real-time updates to Firebase and dashboard.

- Buzzer Alert System
  - Generates alerts for security events and invalid access attempts.

- Power Regulation Circuit
  - Ensures stable voltage supply to ESP32 and peripheral devices.

- Successfully tested locker locking and unlocking operations under multiple user scenarios.
- Implemented door status monitoring using magnetic reed switch sensors.
- Developed independent locker node architecture enabling fault isolation and improved reliability.
- Optimized wiring and hardware layout for future PCB deployment and industrial implementation.

Software & Cloud Infrastructure

Web Portal Development

- Developed responsive web-based locker booking platform.
- Implemented user registration and booking workflow.
- Integrated dynamic locker allocation mechanism.
- Developed session management system with automatic timer control.

Web Portal Features:

- User registration and login.
- Locker booking and allocation.
- PIN generation and validation.
- Session timer monitoring.
- Booking history management.

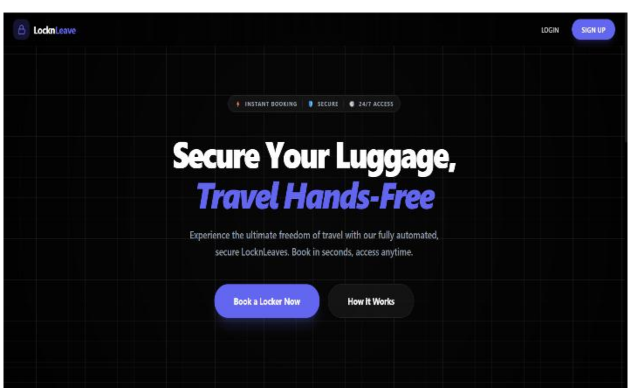

Figure: User Web Portal Home Page

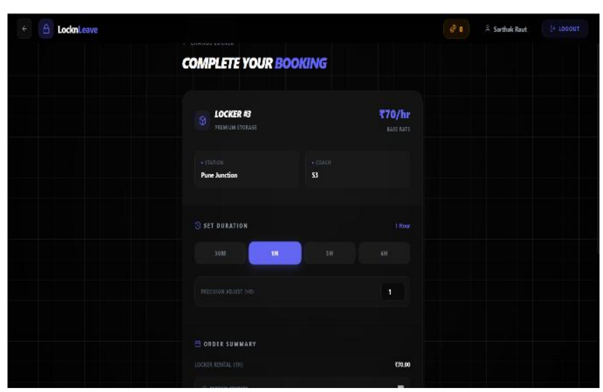

Figure: Locker Booking Interface

Firebase Integration

- Firebase Realtime Database used for:
  - Locker Status Synchronization
  - Session Management
  - User Data Storage
  - Device Communication
- Achieved real-time synchronization between hardware and cloud backend.
- Implemented automatic database updates for locker occupancy and availability.

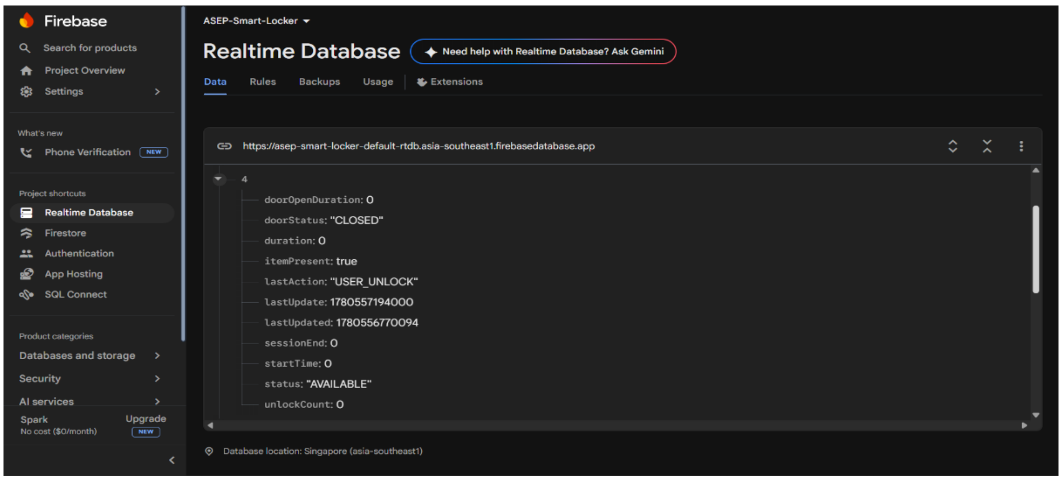

Figure: Firebase Realtime Database Integration

Security & Monitoring Features

- Implemented secure Virtual PIN Authentication System using cloud-based credential verification.
- Developed automatic session expiration mechanism.
- Integrated buzzer alerts for security events.
- Implemented door-open monitoring using reed switch feedback.
- Added emergency administrative override functionality.
- Established secure real-time communication between Firebase cloud services and ESP32 locker controllers.

Multi-Locker Management System

- Designed scalable architecture supporting multiple locker units.
- Developed centralized monitoring dashboard for locker management.
- Real-time visibility of:
  - Available Lockers
  - Occupied Lockers
  - Active Sessions
  - Booking Records
- Architecture supports future deployment in railway stations, airports, bus terminals, and public facilities.

Dashboard Features

- Real-time locker availability monitoring.
- Occupancy status visualization.
- Active session tracking.
- Booking record management.
- Emergency locker control.

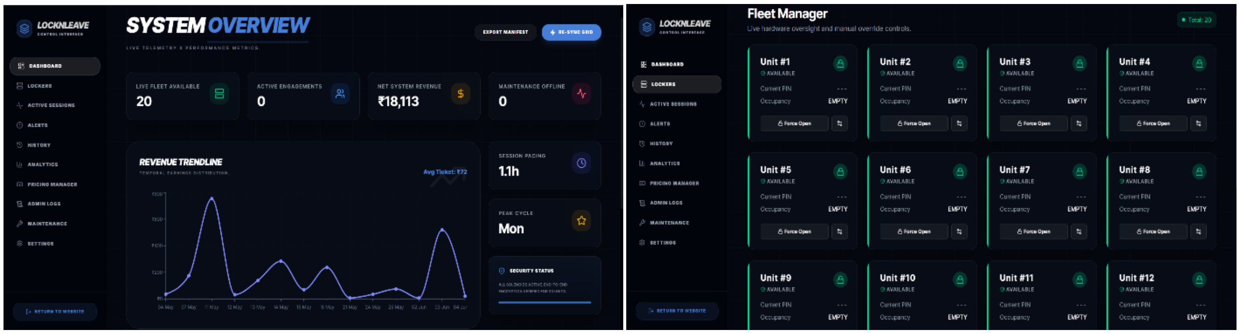

Figure: Centralized Locker Management Dashboard

Testing & Validation

Functional Testing Results

All major system functionalities were successfully validated during testing.

- User booking workflow executed successfully.
- Virtual PIN generation and authentication verified.
- Solenoid lock unlocking and locking operations confirmed.
- Reed switch door status monitoring validated.
- Session expiry and automatic locker release tested successfully.
- Real-time synchronization between Firebase and ESP32 verified.

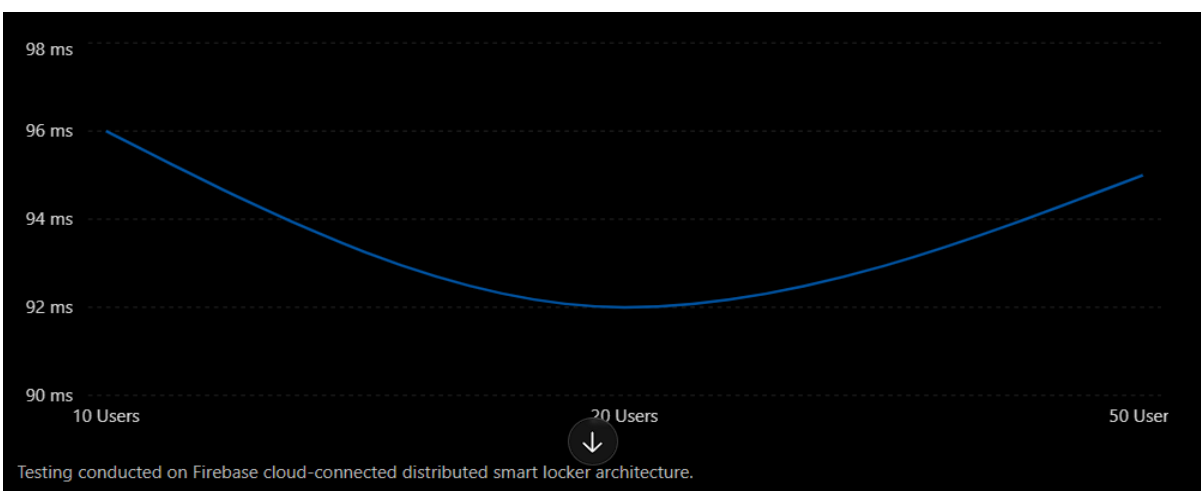

Figure: Functional Testing of Smart Locker System

Load Testing (Apache JMeter)

Concurrent Users	Avg Response Time	Error Rate
10 Users	96 ms	0 %
20 Users	92 ms	0 %
50 Users	95 ms	0 %

The system maintained stable performance under concurrent user loads and achieved a 0% error rate during all test scenarios.

Results Achieved

- Successfully developed and deployed a fully functional IoT-enabled Smart Locker prototype.
- Achieved real-time synchronization between ESP32 hardware and Firebase cloud backend.
- Implemented secure virtual PIN authentication using TFT touchscreen interface.
- Successfully validated door monitoring using reed switch sensors.
- Developed centralized dashboard for monitoring locker status and active sessions.
- Demonstrated stable system performance with 0% error rate during load testing.
- Successfully executed complete workflow from booking to locker access and session completion.

Documentation & Intellectual Property Work

- Completed detailed system architecture documentation.
- Prepared circuit diagrams, hardware architecture diagrams, and software workflow diagrams.
- Developed technical specifications for multi-locker deployment.
- Prepared patent documentation including invention description, system architecture, and operational workflow.

Current Project Status

- Fully functional smart locker prototype completed.
- Hardware and software integration completed.
- Cloud synchronization operational.
- Real-time monitoring dashboard operational.
- Multi-locker architecture finalized.
- System ready for demonstration, evaluation, and future large-scale deployment.

Actual System Image